

GenAI Attitudes, Skills, Use & Perceived Impact - Construct Specifications

GenAI Attitude

Multidimensional feature capturing students' reliance on GenAI for validating their reasoning and generating ideas, as well as their belief that GenAI can serve as an alternative to in-person education.

Survey 1: Sum of (dis)agreement with statements

1. Compared to the situation without using AI tools, thanks to the AI tools ...I attend less classes.
2. AI gives me new ideas I wouldn't think of.
3. I use AI to check or correct my reasoning.
4. I trust it and use it immediately.
5. I check its reasoning or cross-check with other sources, like textbooks.
6. I use it as inspiration but rewrite or rework it completely. (reversed)
7. I compare its approach with mine to learn different methods. (reversed)

Survey 2: Sum of (dis)agreement with statements

8. AI adapts explanations to my learning style better than human teachers.
9. If I miss a lecture/tutorial, AI helps me catch up effectively.
10. Because AI is always available, I sometimes prefer it over attending classes.
11. I think using AI is a skill I need for my future studies/career.
12. AI gave me new perspectives or ideas.

GenAI Use Skill

Sum of reported frequencies of "good GenAI practices" (e.g. prompt engineering, project separation by scope).

Survey 2: Sum of reported frequencies

For each statement, indicate how often you used the following practices. (I didn't know this was possible; Never -- Always)

1. Uploading relevant course materials (PDFs, slides, notebooks)
2. Using specialized AI modes (study, exam, etc.)
3. Specifying difficulty level
4. Configuring/asking for specific output formats, like .docx, .pdf, etc.
5. Keeping separate AI chats per course/topic
6. Keeping my AI chats organized (e.g. into folders/projects by topic or course)
7. Configuring/asking AI to check or challenge my reasoning
8. Configuring/asking AI to ask clarifying questions when my prompt is unclear
9. Verifying AI-generated explanations or solutions with external sources (course materials, textbooks, etc.)
10. Using AI to generate practice questions or quizzes
11. Using AI to plan/schedule study tasks
12. Using AI to help debug code by uploading my code and error messages
13. Using AI to give feedback on clarity of writing/explanations

Study Aid Use

Captures support-oriented interactions such as summarization, explanation, and example generation.

Survey 1: Sum of reported practices

1. To understand new concepts (Q2)
2. To summarize or organize course material (e.g. a “cheat sheet” with formulas) (Q2)
3. To revise topics and check understanding (Q3)
4. To create practice questions or flashcards (Q3)
5. To generate examples or solutions (Q3)
6. To explain exam questions or past paper problems (Q3)
7. I ask AI for explanations (Q4)
8. I use AI to test myself (e.g., by asking it to quiz me) (Q4)
9. I ask AI to show me examples before I try on my own (Q4)
10. Ask for explanations of syntax or logic (Q6)
11. Learn programming by experimenting with AI-generated examples (Q6)

Survey 2: Sum of reported usefulness (Section B)

12. Understanding theory concepts (Theory Exercises)
13. Summarizing lecture material (Theory Exercise)
14. Creating examples to understand concepts (Theory Exercises)
15. Understanding exercises (what the exercise was asking me to do) (Theory Exercises)
16. Understanding how to solve theory exercises (Theory Exercises)
17. Understanding Python syntax and concepts (Programming Exercises)
18. Understanding tasks (what the exercise was asking me to do) (Programming Exercises)
19. Learning programming through experimentation (Programming Exercises)
20. Understanding the dataset (Graded Assignment: Jupyter Notebook 1 - EDA)
21. Understanding the tasks (Graded Assignment: Jupyter Notebook 1 - EDA)
22. Choosing an appropriate analytical approach (e.g. regression model, hypothesis tests, etc.) (Graded Assignment: Research Question & Approach)
23. Justifying why my chosen method was appropriate (Graded Assignment: Research Question & Approach)
24. Understanding Python functions or coding logic (Graded Assignment: Jupyter Notebook 2 - Analysis)

Improve Use

Reflects iterative improvement of student-produced artifacts, including debugging code, enhancing text, and providing feedback.

Survey 1: Sum of reported practices

1. To brainstorm or explore ideas for projects
2. To improve writing or presentations
3. To check or proofread written work (solutions, reports, etc.)
4. I use AI to improve / fix my text, slides, code, ...
5. Debug or optimize code I wrote

Survey 2: Sum of reported usefulness (Section B)

6. Debugging (finding and correcting errors in the code) (Programming Exercises)
7. Checking whether my solution was correct or complete (Programming Exercises)
8. Debugging (Graded Assignment: Jupyter Notebook 1 - EDA)
9. Checking whether my solution was correct or complete (Graded Assignment: Jupyter Notebook 1 - EDA)
10. Brainstorming Research Question (Graded Assignment: Research Question & Approach)
11. Refining the wording of the Research Question (Graded Assignment: Research Question & Approach)
12. Improving written text (Graded Assignment: Research Question & Approach)
13. Debugging errors (Graded Assignment: Jupyter Notebook 2 - Analysis)
14. Improving clarity of my text (Graded Assignment: Poster & Pitch)
15. Preparing talking points (Graded Assignment: Poster & Pitch)
16. Using AI to critique my work (Graded Assignment: Poster & Pitch)

Generate Use

Captures artifact creation from scratch, such as generating code, text, slides, or images directly used in submissions.

Survey 1: Sum of reported practices

1. To generate text for reports or presentations
2. To debug or generate code
3. I use AI-generated material (text, slides, code, ...) directly in my work
4. Generate code from a task description

Survey 2: Sum of reported usefulness (Section B)

5. Generating code (Programming Exercises)
6. Generating code (Graded Assignment: Jupyter Notebook 1 - EDA)
7. Writing observations (Graded Assignment: Jupyter Notebook 1 - EDA)
8. Generating code (Graded Assignment: Jupyter Notebook 2 - Analysis)
9. Interpreting results (Graded Assignment: Jupyter Notebook 2 - Analysis)
10. Generating content for the pitch or poster (Graded Assignment: Poster & Pitch)

Perceived Learning Impact

Students' subjective evaluation of how GenAI use affects their learning in terms of understanding, motivation, time management, and self-confidence.

Survey 1: Sum of (dis)agreement with statements

Compared to the situation without using AI tools, thanks to the AI tools ...

1. I better understand the course material.
2. I think more critically about what I learn.
3. I can solve problems on my own better.
4. I manage my study time more effectively.
5. I feel more motivated to study deeply.

Survey 2: Sum of (dis)agreement with statements / reported frequencies

6. AI makes understanding course material easier.

7. AI made me feel more confident handling the course workload.
8. AI clarified concepts I struggled with.
9. AI sometimes made me skip steps instead of learning them. (reversed)
10. AI helped me solve coding problems.
11. AI made me less confident about my abilities. (reversed)
12. AI made me feel dependent on it for completing tasks. (reversed)
13. AI helped me discover flaws in my reasoning.
14. AI caused me to spend more time than needed. (reversed)
15. AI motivated me to perform a deep analysis.

Belief in GenAI-as-Teacher

Reflects the belief, or even preference, that GenAI can replace traditional education.

Survey 1: Sum of (dis)agreement with statements

1. Compared to the situation without using AI tools, thanks to the AI tools ...I attend less classes.

Survey 2: Sum of (dis)agreement with statements

2. AI adapts explanations to my learning style better than human teachers.
3. If I miss a lecture/tutorial, AI helps me catch up effectively.
4. Because AI is always available, I sometimes prefer it over attending classes.
5. I sometimes find AI feedback difficult to understand or apply. (reversed)

GenAI Attitude Re-specified

Following the post-hoc analysis in which GenAI-as-Teacher items were removed from the GenAI Attitude construct, only openness items are retained, as trust items no longer correlate sufficiently. Items recovered from the item bank are likewise a measure of openness.

Survey 1: Sum of (dis)agreement with statements

1. AI gives me new ideas I wouldn't think of.
2. I use AI to check or correct my reasoning.

Survey 2: Sum of (dis)agreement with statements

3. I think using AI is a skill I need for my future studies/career.
4. AI gave me new perspectives or ideas.
5. I feel confident I could apply the data analysis skills taught in this course in future projects using AI.
6. I feel confident that I know how to use AI effectively for academic learning.
7. I think using AI is a skill I need for my future studies/career.